GARNET versus FEMFFUSION: time to accuracy

IAEA-3D, 3-D Langenbuch and the hex-homogenized HP-MR, steady and transient

GARNET performance report

November 2026

Summary

This report times GARNET against FEMFFUSION on the same multigroup diffusion problems, solved to the same accuracy:

- IAEA-3D;
- the 3-D Langenbuch core at $t = 0$;
- the 11-group hex-homogenized HP-MR;
- a $+0.5$ \$ reactivity insertion in the HP-MR.

Both codes run on one core of the same AMD EPYC 7742 node, and GARNET also on one GH200 GPU. Every run is scored against a common reference. For each accuracy target, the comparison takes each platform's cheapest measured run that meets it (Table 2).

Findings.

- **Steady state, one core: FEMFFUSION is faster.** At 10 pcm in k and 1% in every node power, its degree-3 elements need 8 to 60 times fewer unknowns than GARNET's second-order finite volumes.
  - HP-MR: 89 s for FEMFFUSION against 667 s for GARNET.
  - Langenbuch: 45 s against 66 s.
  - IAEA-3D: 658 s; GARNET needs $r = 24$ (127 million unknowns), an estimated 2.6 h on one core, which was not run.
  - 1 pcm and 0.1%: only FEMFFUSION reaches it (Langenbuch 434 s, HP-MR 124 s, IAEA-3D 10,822 s).
- **Steady state, GH200: GARNET is fastest on all three cores.** HP-MR in 2.4 s, 37 times faster than FEMFFUSION on one core; Langenbuch in 0.35 s, 130 times; IAEA-3D in 33.5 s, 20 times.
- **HP-MR: GARNET's Wielandt shift.** Its HP-MR rows use the new Wielandt-shifted eigenvalue iteration (classic fixed shift, $\delta k = 0.3$). Each outer iteration then solves the whole 11-group system, upscatter included. Outer iterations drop from about 250 to 10, and solve times fall 2.2 to 3.5 times on one core and on the GH200 (Section 4). The two-group cores keep the power iteration: there the shift is 1.4 to 1.6 times slower on the GH200.
- **IAEA-3D: GARNET's node powers need fine meshes.** At $r = 16$ (37.5 million unknowns, 1.25 cm cells) its k is within 10 pcm, but its node powers are still up to 2.1% off: a smooth tilt over the core, because the cells barely resolve the 2 cm thermal diffusion length. It reaches 1% at $r = 24$ (127 million unknowns), and its mesh limit matches FEMFFUSION's reference to 0.01 pcm and 0.05%: both codes converge to the same solution.
- **Transient: GARNET is faster on the same core.** To keep the power error below 10^{-3}:

  - FEMFFUSION: 1262 s on one core;
  - GARNET: 127 s on one core (10 times faster) and 17 s on the GH200 (76 times).

  At 10^{-2} the two codes are even on one core (107 and 100 s).

- **Why the transient differs: time integration.** GARNET's BDF2 is second order; FEMFFUSION's two usable schemes are first order. FEMFFUSION's default scheme, Implicit-Exponential, does not hold the unperturbed state on this input.

Scope.

- FEMFFUSION runs on one core only: its MPI path does not work yet (Section 1). The GH200 figures are therefore against one core, not node against node.
- The transient reference is GARNET's extrapolated GH200 solution. FEMFFUSION's own runs converge onto it independently.
- The HP-MR drums are withdrawn. A smeared B_4C drum is black to thermal neutrons, and neither code resolves it.
- Posing the problems identically in both codes uncovered five FEMFFUSION issues, listed in Section 6.

1 Codes, builds and hardware

FEMFFUSION. Commit `main@1611f90` of `https://github.com/Zonni/FEMFFUSION`, unmodified. It was built for this report against the library versions its README documents. All three libraries were compiled from source with `-O3 -march=znver2`, so deal.II's matrix-free kernels use 4-wide AVX2 vectors:

- PETSc 3.19.6 (real scalars, 32-bit indices);
- SLEPc 3.19.2;
- deal.II 9.5.2 (Release, with PETSc and SLEPc).

The toolchain is gcc 12.3, Open MPI 5.0.8 and OpenBLAS 0.3.30. The build recipe is `slurm/build_femffusion.sh`. Its own test suite (`femffusion.exe -t`) passes: "ALL TEST PASSED".

FEMFFUSION runs on *one core*. Its data structures are written for MPI (`parallel::shared::Triangulation`, PETSc MPI matrices), but several places assume a single process:

- `MPI_Bcast` of a vector object's address;
- a local index in `copy_to_Vector`, used by every matrix-free product;
- the shell-matrix sizes of `slepc_2g`;
- the Newton and multilevel paths, which refuse more than one process.

Threads are capped at one (`MPI_InitFinalize(..., 1)`). No upstream branch or fork carries MPI work. Repairing it is deferred, so a full-node FEMFFUSION column is not part of this report.

GARNET. Branch `femffusion-perf-report` (from `main@c81ee74`), NumPy/SciPy on the CPU and CuPy on the GPU, double precision, fused kernels on. The HP-MR steady runs with the Wielandt shift come from branch `wielandt-shift`, which adds it to `femffusion-perf-report@dd03736`. On the CPU it is single-threaded (BLAS threads are set to 1).

Hardware.

- **CPU:** one core of an exclusive merlin7 node, dual AMD EPYC 7742 (Zen 2, 2×64 cores, NPS4, 512 GB). Concurrent single-core runs are pinned one per NUMA domain. Both codes' CPU runs use the same node type. The reference-only runs (Section 4) shared a node with other users' jobs; their times are not used.
- **GPU:** one NVIDIA GH200 (96 GB HBM3) on gmerlin7.

Speed-ups are quoted against one core. They are not node-to-node comparisons.

2 Problems: the same equations in both codes

All four problems are multigroup diffusion on homogenized regions, and each is posed to both codes with identical data. Discretization is the only difference. `FEMFFUSION` uses continuous-Galerkin finite elements of degree p on hexahedra. `GARNET` uses second-order cell-centred finite volumes on Cartesian boxes or triangular prisms. Both therefore converge to the same continuous solution.

IAEA-3D. The two-group benchmark (ANL-7416 Suppl. 2) on a $17 \times 17 \times 19$ lattice of 20 cm nodes. `FEMFFUSION` runs its own input (`examples/3D_IAEA`). That input applies the albedo condition $D\,\partial\phi/\partial n = -0.4695\,\phi$ on every face of the full core, and the corners of the octagonal core are holes. `GARNET`'s `build_iaea` carries the same data. Its albedo is the same Robin coefficient: `FEMFFUSION` `matrix_operators_petsc.cc` adds $\alpha \int \phi\, v$ on boundary faces, and `GARNET` passes α to its Robin face term. The published $k = 1.02903$ assumes a different boundary condition. It is quoted for context only and is not used as a reference.

3-D Langenbuch at $t = 0$. The two-group LMW core ($11 \times 11 \times 10$ nodes of 20 cm) with zero-flux boundaries. Bank 1 is half inserted, with its tips at 100 cm on a node face, and bank 2 is withdrawn, so no node is partly rodded. `FEMFFUSION` runs `examples/3D_Langenbuch` with its `.bar` file at the initial positions. Two discrepancies in `GARNET`'s builder were found and removed before any timing:

- `FEMFFUSION`'s map excises the four corner columns. `GARNET` had filled them with reflector. `build_langenbuch(excise_corners=True)` now poses the staircase zero-flux boundary instead; the default is unchanged.
- `GARNET`'s cross sections had been transcribed with one digit fewer than the `.xsec` file (a relative difference of about 10^{-8}). They now carry the file's digits.

Input identity, IAEA-3D and Langenbuch. `tests/validation/test_femffusion_equivalence.py` rebuilds `FEMFFUSION`'s node map from the `.prm Geometry_Points` and the `.xsec PLANE` blocks, and applies the bars at their initial heights. It then checks node by node that `GARNET`'s material map, active mask, cross sections and boundary conditions are the same. `GARNET`'s first array axis is `FEMFFUSION`'s y, so `GARNET`'s maps are `FEMFFUSION`'s transposed in plan. The two models are mirror images across the core diagonal: k is identical, and node powers are compared after the transpose.

HP-MR, hex-homogenized (11 groups). The ANL/INL heat-pipe microreactor: 30 fuel hexagons, a central cell, 12 beryllium reflector hexagons and 12 control drums, on a 26.752 cm lattice. The core is 200 cm high, with 20 cm beryllium axial reflectors. Cross sections are the 11-group ENDF/B-VIII.0 library of the INL Virtual Test Bed, with the fuel assembly flat-flux homogenized. Each hexagon is one material in each axial zone. The drums are withdrawn: a drum hexagon is the library's drum beryllium, with no B_4C (Section 2 explains why). The boundary is vacuum (Marshak, $\alpha = 1/2$) on every face in both codes. `FEMFFUSION`'s input is written by `tools/export_femffusion.py`:

- a Gmsh mesh of three quadrilaterals per hexagon, built from `GARNET`'s own site coordinates;

- an XML material file with one mix per hexagon and plane: 55 hexagons × 10 planes of 20 cm;
- the kinetics in a `PREC_Filename` file.

The XML writer round-trips through `GARNET`'s reader exactly. The only exception is D, which passes through $\Sigma_{tr} = 1/(3D)$ as in `FEMFFUSION` and so loses one ulp.

HP-MR transient: external reactivity insertion. From the critical initial state, every fissile $\nu\Sigma_f$ is multiplied by $1 + \varepsilon(t)$. ε rises linearly from 0 to $\varepsilon_1 = \rho/(1-\rho)$ over $[0, 2]$ s, with $\rho = 0.5\,\beta = 340.1$ pcm. Scaling all production scales k, so the final state is exactly $+0.5$ \$ statically on any mesh. Σ_f, the power weight, is unchanged.

The kinetics are the library's 11 group speeds and six precursor families ($\beta = 680.2$ pcm), with the delayed spectrum set equal to χ in both codes, and there is no feedback. `FEMFFUSION` runs this through `Type_Perturbation = Read_XML_File`, which interpolates every cross section linearly from the initial to a final XML over $[0, T_{\text{end}}]$; here the two files differ only in `NuSigF`. `GARNET`'s `problem_at` applies the same linear scaling.

Three `FEMFFUSION` transient findings. All three were found with a null transient (final XML equal to the initial one), which must keep $P = 1$.

1. With the kinetics in a `PREC_Filename` file (the format of `FEMFFUSION`'s C5G7-TD example), an XML material keeps a zero prompt spectrum. `FEMFFUSION` stores the prompt and delayed spectra of an XML material only when it has `ChiD`, and the unperturbed transient collapsed. The kinetics are therefore written into the XML.
2. The default Implicit-Exponential scheme takes the decay constants and delayed spectra of material 0 for the whole core (`get_delayed_decay_constant(0, p)`). Our mix 0 is a non-fissile axial-reflector cell, so every non-fissile mix carries the global λ and the fissile χ with $\beta = 0$; all fissile mixes share one set of delayed data. This changes nothing physically.
3. Even so, Implicit-Exponential does not hold the null transient: P drops by 3.9% in a 0.1 s step and by 0.2% in a 0.01 s step, and keeps drifting. Both semi-implicit schemes (Semi-Implicit-Euler and Semi-Implicit-Exponential) hold $P = 1.0000$. The comparison uses those two.

Why the drums are withdrawn. The model was first set up with each drum hexagon homogenized by area: the drum's 90° B_4C arc covers $f_0 = 3.2\%$ of the hexagon, with Σ_t mixed and $D = 1/(3\Sigma_t)$ (`absorber="smeared"`). The result is a hexagon that is black to thermal neutrons: $\Sigma_a \approx 10$ cm^{-1}, $D \approx 0.03$ cm, a diffusion length of about 0.5 mm. Neither code converges on it at any practical mesh:

- `FEMFFUSION` FE1 to FE3 levels off near $k = 1.0347$;
- `GARNET` came down from 1.0553 ($r = 2$) through 1.0494 ($r = 4$) to 1.0466 ($r = 6$, 7.8 million unknowns), roughly first order and still 1200 pcm above.

With the drums made pure beryllium, the two codes agree to a few pcm at the same meshes. That confirms the geometry, boundary conditions and cross-section semantics. Such a drum would be a benchmark of resolving an unphysical sub-millimetre boundary layer, not of discretization cost.

The same smearing rules out a drum-rotation transient: a linear ramp of the smeared B_4C fraction inserts almost all of its reactivity in its last few percent, because the hexagon stays black until the fraction falls below about 10% of nominal. The transient is therefore an external reactivity insertion.

3 Method

Time to accuracy. Each code runs a ladder of discretizations:

- `FEMFFUSION`: degree 2 and degree 3 elements, each at successive uniform refinements;
- `GARNET`: successive cell refinements r (and, for the HP-MR, axial layers n_z that keep the prisms roughly isotropic).

Every run is scored against the same reference. A table entry is the cheapest *measured* run that meets a target; nothing is interpolated between runs. The steady targets are $|\Delta k| \leq 10$ pcm with every node power within 1%, and $|\Delta k| \leq 1$ pcm with 0.1%. The transient targets are a maximum $|P/P_{\mathrm{ref}} - 1|$ of 10^{-2} and of 10^{-3}.

Tolerances. Each code uses the loosest iteration tolerance whose iteration error stays below 0.1 pcm in k and 10^{-4} in node power. That threshold is well below the discretization errors being compared:

- `GARNET`: `tol_k` $= 10^{-7}$ and `tol_source` $= 10^{-6}$, measured against $10^{-11}/10^{-10}$ on IAEA, Langenbuch and the first, smeared-drum HP-MR (at most 0.007 pcm and 6.6×10^{-5}). On the drums-withdrawn HP-MR compared here ($r = 2$, $n_z = 30$) the power iteration is 0.006 pcm and 2.4×10^{-6} off, and the Wielandt shift 0.003 pcm and 1.8×10^{-7};
- `FEMFFUSION`: `EPS_Tolerance` $= 10^{-6}$ for `slepc_2g`, `gd` and `bifpam`, measured against 10^{-10} or 10^{-11} (below 0.05 pcm and 1.7×10^{-5}).

For the transients, `FEMFFUSION`'s linear tolerance `KSP_Tolerance` $= 10^{-6}$ changes $P(2\,\mathrm{s})$ by 5×10^{-6} relative to 10^{-9}. `GARNET`'s within-step tolerance is 10^{-7} for fixed steps and 10^{-8} for adaptive steps.

Configuration. Both codes were tuned on mid-size meshes, on the same hardware as the production runs. The full sweeps are in `data/perf/*tune*`; the chosen settings are:

- `FEMFFUSION`, steady: `non_diagonal` storage with `P_Init` on; `slepc_2g` for IAEA, `bifpam` for Langenbuch, and for the HP-MR `gd` at degree 2 and `bifpam` at degree 3.
- `FEMFFUSION`, transient: `KSP_Tolerance` 10^{-6} and bad-Broyden preconditioner updates; three to four times faster than the fixed preconditioner.
- `GARNET`, CPU: Neumann preconditioner of degree 1/0/0 and Anderson depth 4/4/8 for IAEA/Langenbuch/HP-MR.
- `GARNET`, GH200: degree 2/2/1 and depth 4/4/8.
- `GARNET`, HP-MR steady: the fixed Wielandt shift $\delta k = 0.3$ (Section 4) with the defaults of its multigroup solve. Only the GH200 run at $r = 12$ shortens the FGMRES restart from 20 to 8, so that the basis fits in the 96 GB.
- `GARNET` transients: BDF2 with the fixed-point step for fixed steps; the monolithic step for adaptive BDF, which supports only that.

Two `FEMFFUSION` options were excluded for errors, not for speed:

- `full_matrixfree` with `gd` or `slepc_2g` crashes (signal 11);
- `full_allocated` with `gd` or `bifpam` on the refined Langenbuch core returns a wrong eigenvalue ($k = 1.702$ at FE2 $r = 1$ and 1.380 at FE3 $r = 1$, against 0.9995).

`power_it` was dropped because it exceeds 300 s where the others take seconds.

Cost. Two times are reported per run.

- *Solve* is each code's own timed computation. For `GARNET` it is the device-synchronized `solve_seconds`; for a transient that is the initial eigen-solve plus the march. For `FEMFFUSION` it is the process CPU time from "Matrices assembled" to the eigenvalue, which includes the degree-1 `P_Init` pre-solve; for a transient, its transient CPU time is added. `FEMFFUSION` runs single-threaded, so its CPU time is its wall time.
- *Wall* is time to solution from a cold start. For `FEMFFUSION` it is the whole process. For `GARNET` it is the builder, the solver construction and the solve, plus, on the GPU only, one warm-up run. That is an upper bound on the CuPy kernel compilation, since the warm-up also does two outer iterations or time steps. NumPy compiles nothing.

Steady runs under a few minutes are the best of three; longer ones are single. The concurrency control re-times runs alone on an otherwise idle node: within 6% of the concurrent runs.

Power. `FEMFFUSION`'s node and total powers are Σ_f-weighted. `GARNET`'s records use the same weight: $\kappa\Sigma_f$, written to `FEMFFUSION` as $\Sigma_f = \kappa\Sigma_f/(200\text{ MeV})$. Node powers are normalized to the core-volume average power density, as `FEMFFUSION` does. `GARNET`'s `TransientResult.power` is the $\nu\Sigma_f$-weighted fission source, which carries the inserted $1 + \varepsilon(t)$. The comparison therefore converts it to the $\kappa\Sigma_f$-weighted rate. The spectral part of that conversion, the ratio of κ- to ν-weighted rates, moves by less than 4×10^{-8} over the transient.

4 Steady state

References. Table 1 lists the reference of each case, how it was obtained, and the independent solutions that check it. The primary reference is the candidate with the smallest own uncertainty:

- Langenbuch: the three-level limit of `FEMFFUSION`'s degree-3 refinements (observed order 3.3), confirmed by degree 4 and 5 to 0.34 pcm;
- HP-MR: `FEMFFUSION`'s finest run, FE4 $r = 1$ (9.6 million unknowns). FE2 $r = 2$ (the same unknowns) and FE3 $r = 1$ are 0.4 pcm below it and within 0.01% in every node, and degree 4 and 5 at $r = 0$ are within 0.6 pcm; `GARNET`'s mesh limit (order 2.2) is 0.5 pcm away;
- IAEA-3D: the same degree-3 limit, confirmed by `FEMFFUSION`'s degree-2 limit to 0.07 pcm (see the paragraph on IAEA below).

Where `GARNET`'s sequence is not yet asymptotic, its limit is listed but not used. On Langenbuch it shows an order of 0.9 against a nominal 2: the error there is dominated by the re-entrant corners of the staircase boundary. A run that *is* the reference is not a comparison row.

Time to accuracy. Table 2 gives, for each target, the cheapest measured run of each platform. Figures 1–3 show every run, with each platform's Pareto front, and Tables 4–6 list them all.

What the steady results show.

- **Per core, `FEMFFUSION`'s high-order elements win, often by an order of magnitude.** On the HP-MR at 10 pcm and 1%, `FEMFFUSION` needs 89 s (FE3 $r = 0$). `GARNET` on one core needs 667 s ($r = 6$, 10.5 million unknowns, with the Wielandt shift below): 7.5 times slower. On Langenbuch the two are comparable at 10 pcm (45 against 66 s). `GARNET`'s cost per unknown is lower (Section 6), but its second-order finite volumes need far more unknowns for the same error, above all for the node powers.

Case	Reference	Value	Uncertainty
IAEA-3D	FEMFFUSION FE3, refinements 0/1/2 (three-level limit, nominal order 3: last step within the printed digits)	$k = 1.0290729$	0.10 pcm
	check: FEMFFUSION FE2 limit (r = 1, 2, 3)	1.0290736	+0.07 pcm
	check: GARNET GH200 limit	1.0290729	+0.01 pcm
3-D Langenbuch ($t = 0$)	FEMFFUSION FE3, refinements 0/1/2 (three-level limit, observed order 3.30)	$k = 0.9994838$	0.34 pcm
	check: FEMFFUSION FE4 r1	0.9994872	+0.34 pcm
	check: FEMFFUSION FE5 r1	0.9994854	+0.16 pcm
	check: FEMFFUSION FE4 r2	0.9994853	+0.15 pcm
	check: FEMFFUSION FE2 limit (r = 1, 2, 3)	0.9994830	-0.08 pcm
	check: GARNET GH200 limit	0.9995353	+5.16 pcm
HP-MR 3-D (hex-homogenized, 11 groups)	FEMFFUSION finest run, FE4 r1 (9,602,307 DoFs) (spread of the three finest FEMFFUSION runs)	$k = 1.1186400$	0.60 pcm
	check: FEMFFUSION FE4 r0	1.1186340	-0.60 pcm
	check: FEMFFUSION FE5 r0	1.1186380	-0.20 pcm
	check: GARNET GH200 limit	1.1186449	+0.49 pcm
HP-MR 3-D, +0.5 $ insertion	GARNET GH200, BDF2 dt = 0.025/0.0125 s extrapolated in time, then in space over (r, nz) = ((4, 50), (6, 80), (8, 100))	$P(2\,\mathrm{s}) = 2.52056$	8e-05

Table 1: References and their checks. Uncertainty: the largest of the extrapolation step, the spread of the independent high-degree solutions, and (for an asymptotic series) half the cross-code difference.

Case	Target	FEMFFUSION best (s)	GARNET CPU (s)	GARNET GH200 (s)	FEMFFUSION/CPU	FEMFFUSION/GH200
IAEA-3D	10 pcm, 1%	658	–	33.5	–	20
	1 pcm, 0.1%	10822	–	–	–	–
3-D Langenbuch ($t = 0$)	10 pcm, 1%	44.9	66.1	0.35	0.68	130
	1 pcm, 0.1%	434	–	–	–	–
HP-MR 3-D (hex-homogenized, 11 groups)	10 pcm, 1%	89.3	667	2.38	0.13	37
	1 pcm, 0.1%	124	–	–	–	–
HP-MR 3-D, +0.5 $ insertion	0.01	107	99.9	12.1	1.1	8.9
	0.001	1262	127	16.7	10	76

Table 2: Time to accuracy (solve time, s): the cheapest run of each platform that meets the target ($|\Delta k|$ and the largest node-power error; for the transient, the largest power error over the run), and the ratio of FEMFFUSION's better degree to GARNET. "–": no measured run meets the target.

- **GARNET's node powers converge slowly on IAEA.** At $r = 16$ (37.5 million unknowns) k is within 10 pcm, but the largest node-power error is still 2.1%. The error is a smooth tilt over the whole core, not a local defect: the power is up to 0.9% high at the centre and 1.4% low at the radial edge. The worst node, at the edge of the top fuel layer and a sixth of the average power, is where the tilt is most negative. Every node converges at the same rate, $h^{1.8}$ from $r = 8$ to 16 (local order 1.6, 1.75, 1.85 with refinement). Extrapolated at that rate, GARNET's $r = 8$ and 16 agree with the reference to 0.06% in every node, so the two codes solve the same problem; the cause is resolution. The thermal diffusion length is 2.2 cm in the fuel and 1.8 cm in the rodded fuel, so the thermal flux changes on a 2 cm scale at every fuel–reflector and rod interface. The $r = 16$ cells are 1.25 cm, and second-order finite volumes are not yet asymptotic there; FEMFFUSION's cubic elements resolve those gradients within each element. The finer GH200 runs confirm it: from $r = 16$ to 28 the local order reaches 2 (1.99, 1.90, 2.00), and $r = 24$ (0.83 cm cells, 127 million unknowns) is the first mesh within 1% (0.95%; $r = 28$: 0.70%). On Langenbuch the worst node, in the same position, is 0.56% off at $r = 16$.
- **On the GH200, GARNET reaches the 10 pcm, 1% target first on all three cores:** 2.4 s on the HP-MR (37 times faster than FEMFFUSION on one core), 0.35 s on Langenbuch (130 times) and 33.5 s on IAEA (20 times). Its 1 pcm, 0.1% target is out of reach of the meshes that fit (HP-MR $r = 12$, 84 million unknowns, is 1.5 pcm and 0.15% off; IAEA $r = 28$, 201 million, 1.7 pcm and 0.70%). That target is close to the references' own resolution (0.1 to 0.6 pcm), so it separates the codes only coarsely.

HP-MR: Wielandt-shifted iteration. GARNET's HP-MR rows use its Wielandt-shifted eigenvalue iteration. Each outer iteration solves $(M - F/k_s)\Psi = (1/k - 1/k_s)F\Psi_{\text{old}}$ over all 11 groups at once by FGMRES, preconditioned by a sweep over the groups. Fission and scattering, upscatter included, are resolved in every outer. The plain power iteration sweeps the groups once per outer and leaves the strong thermal upscatter of this library to the outer loop: about 250 outers at every mesh. Table 3 compares the two shift modes, on the same node types, tolerances and preconditioners as the power iteration they replace:

- **fixed** (classic Wielandt): $k_s = k + 0.3$, frozen after the first outer;
- **adaptive**: $k_s = k + \delta$ at every outer, with δ steered so a shifted solve costs about six FGMRES iterations.

The fixed shift is 9–25% faster on every mesh, CPU and GH200, and is the one used in the comparison rows. It is 2.2–3.5 times faster than the power iteration, and moves k by at most 0.005 pcm.

On the two-group cores the power iteration with Anderson acceleration stays faster, so their rows are unchanged. Measured on the GH200 at the meshes that decide Table 2 (best of three), it takes 0.33 s on Langenbuch $r = 12$. The fixed shift takes 0.45 s, $\delta k = 0.1$ 0.47 s and the adaptive shift 0.53 s. On IAEA $r = 24$ the times are 34.1 s against 47.0, 52.6 and 50.8 s: the shift is 1.4–1.6 times slower on both. It halves the outers (51 to 27 on IAEA), but its shifted solves need more within-group iterations than the outers save (4263 against 3197). Its FGMRES basis also nearly doubles the GPU memory (76 against 40 GB at IAEA $r = 24$). Without upscatter, the coupled solve has little to resolve that Anderson does not.

			power iteration		adaptive shift		fixed shift			
Platform	Mesh	Unknowns	outers	s	outers	s	outers	s	speed-up	$\lvert\Delta k\rvert$ (pcm)
GARNET CPU	r2 z30	435,600	253	18.2	12	9.37	10	8.27	2.2	0.003
GARNET CPU	r3 z40	1,306,800	252	76.4	12	36.1	10	31	2.5	0.002
GARNET CPU	r4 z50	2,904,000	251	241	12	110	10	94.7	2.5	0.001
GARNET CPU	r6 z80	10,454,400	250	1995	12	770	10	667	3.0	0.002
GARNET GH200	r2 z30	435,600	254	3.01	12	1.15	10	0.977	3.1	0.005
GARNET GH200	r3 z40	1,306,800	252	3.81	11	1.28	10	1.17	3.2	0.001
GARNET GH200	r4 z50	2,904,000	252	4.78	12	1.7	10	1.36	3.5	0.001
GARNET GH200	r6 z80	10,454,400	251	6.96	12	2.72	10	2.38	2.9	0.001
GARNET GH200	r8 z100	23,232,000	250	14.5	12	6	10	5.23	2.8	0.002
GARNET GH200	r12 z160	83,635,200	250	68.2	12	26.7	10	23.2	2.9	0.003

Table 3: HP-MR steady, GARNET: power iteration (Anderson depth 8) against the adaptive and fixed Wielandt shift, $\delta k = 0.3$, best of three solve seconds except CPU $r = 6$ (single runs). Speed-up: power iteration over fixed shift; $|\Delta k|$: fixed shift against power iteration.

IAEA-3D. The reference is FEMFFUSION's degree-3 limit, as on Langenbuch: FE3 $r = 0$, 1, 2 (16.1 million unknowns at $r = 2$, 4.9 h on a shared node). Its k steps are 6.4 and 0.1 pcm. The second is one printed digit, since FEMFFUSION prints k to 0.1 pcm here, so no order can be fitted. The limit is taken at the nominal order 3, and its uncertainty is one printed digit, 0.1 pcm. The node powers converge cleanly: each refinement shrinks the step about 12-fold (0.34% at most from $r = 1$ to 2), and the extrapolation moves no node by more than 0.05%. FEMFFUSION's FE2 series ($r = 1$, 2, 3, observed order 4.1) is an independent check: its limit is 0.07 pcm away, and FE2 $r = 3$ is within 0.08% in every node.

GARNET's GH200 mesh limit is a second independent check. From $r = 16$, 24, 28 (observed order 1.98) it is 0.01 pcm from the reference and within 0.05% in every node: the two codes converge to the same solution. From $r = 8$, 12, 16 (order 1.57), the provisional reference before FE3 $r = 2$ was in, it was 1.4 pcm off and up to 0.54% in the node powers, because those meshes are not yet asymptotic.

- FEMFFUSION meets 10 pcm and 1% at FE2 $r = 2$ (4.8 million unknowns, 658 s) and at FE3 $r = 1$ (2.0 million, 665 s).
- GARNET meets it on the GH200 at $r = 24$ (127 million unknowns, 40 GB): 2.3 pcm and 0.95% in 33.5 s, 20 times faster than FEMFFUSION. No one-core run meets it. The finest, $r = 16$ (2120 s), is

5.1 pcm and 2.1% off; $r = 24$ would take about 2.6 h on one core (the CPU runs take about 280 times the GH200 time at $r = 12$ and 16). The small Δk at $r = 2$ (-1.5 pcm) is a cancellation of errors: its node powers are 31% off.

- At 1 pcm and 0.1%, only FE2 $r = 3$ qualifies (38 million unknowns, 10,822 s on one core: 0.11 pcm and 0.08%). FE2 $r = 2$ and FE3 $r = 1$ meet the k part but miss the node powers (0.55 and 0.39%). FE3 $r = 2$ is part of the reference and is not a comparison row. `GARNET`'s finest, $r = 28$ on the GH200 (201 million unknowns, 63 GB, 64 s), is 1.7 pcm and 0.70% off.

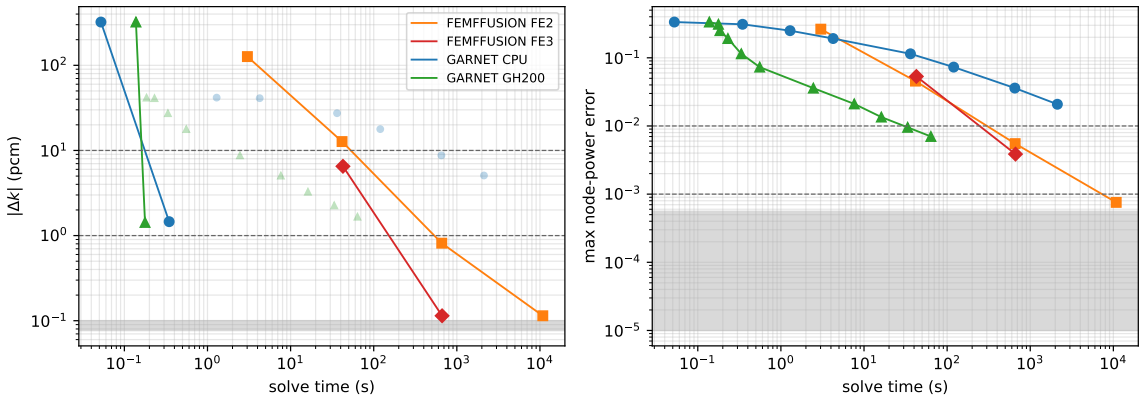


Figure 1: IAEA-3D: $|\Delta k|$ and the largest node-power error against solve time. Faint: every run; line: each platform's Pareto front; dashed: the targets; grey: below the reference's uncertainty (hollow markers sit there).

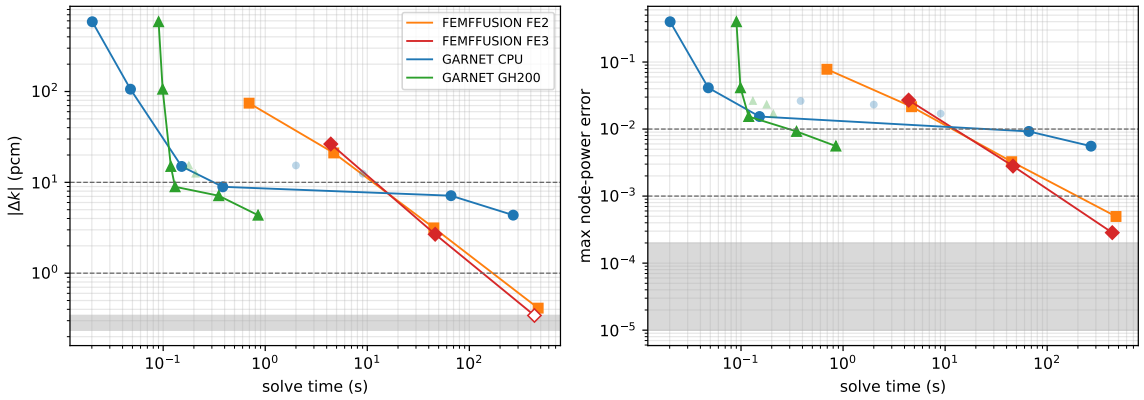


Figure 2: 3-D Langenbuch at $t = 0$, as Figure 1.

5 HP-MR transient: +0.5 $ external reactivity insertion

Reference. The reference is $P(t)$ from `GARNET` on the GH200. Fixed-step BDF2 at $\Delta t = 0.025$ and 0.0125 s is extrapolated in time (order 2), then in space over $(r, n_z) = (4, 50)$, $(6, 80)$ and $(8, 100)$. The two corrections are 1.5×10^{-4} and 6.8×10^{-5} relative, so the reference is good to about 10^{-4}.

The ($r = 12$, $n_z = 160$) mesh would have tightened it, but it does not fit in the GH200's 96 GB: the transient's precursor fields and BDF history raised the allocation past 100 GB, where the eigen-solve on the same mesh needs 74 GB. `FEMFFUSION`'s own runs converge onto this reference independently: its Semi-Implicit-Exponential error falls from 1.1×10^{-3} to 4.5×10^{-4} as Δt halves to 0.003125 s. $P(2\,\text{s}) = 2.5206$.

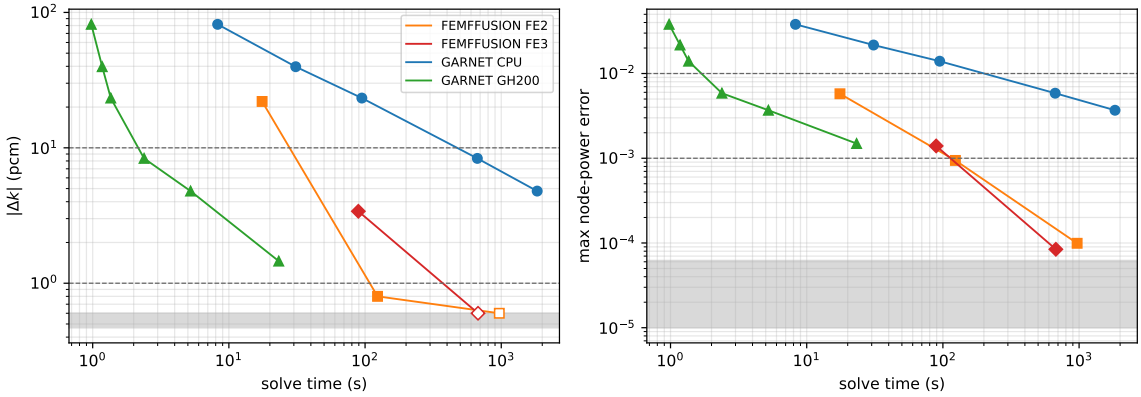


Figure 3: HP-MR, hex-homogenized, 11 groups, as Figure 1.

Time to accuracy. See Table 2 (last rows) and Figure 4.

- 10^{-2}**:** the two codes are even on one core (`FEMFFUSION` 107 s, `GARNET` 100 s). The GH200 takes 12 s.
- 10^{-3}**:** `GARNET` on one core needs 127 s ($r = 2$, BDF2, $\Delta t = 0.1$ s) and `FEMFFUSION` 1262 s (FE2 $r = 0$, $\Delta t = 0.003125$ s), a factor of 10. On the GH200 it is 17 s, 76 times faster than `FEMFFUSION`.

The `GARNET` transients were not rerun with the Wielandt shift (Section 4). Their initial eigen-solve is the power iteration, the same solve as the steady $r = 2$ run: 18.2 s on one core and 3.0 s on the GH200, against 8.3 and 1.0 s with the shift. At 10^{-3} the shift would therefore save about 10 s of the 127 s on one core and 2 s of the 17 s on the GH200; the comparison with `FEMFFUSION` does not change.

The difference is the time integration, not the space discretization. `FEMFFUSION`'s two working schemes are first order. Its default Implicit-Exponential scheme does not hold the unperturbed state on this input (Section 2). `GARNET`'s BDF2 is second order. `FEMFFUSION`'s spatial error is small at every mesh: FE2 $r = 0$ and FE3 $r = 0$ differ by less than 10^{-3}. Its error is therefore almost all time error, and a 10^{-3} target costs it 640 steps.

`GARNET` step control. Down to about 10^{-3} the front is fixed-step BDF2 with the fixed-point step. Below that it is adaptive BDF with automatic order up to BDF4 and an error per step of 10^{-3}: 26 steps (6 rejected) reach 3.2–3.3×10^{-4} on every mesh, in 25 s at $r = 2$ on the GH200.

Tighter error tolerances do not help. Both adaptive variants stall at 2.3–3×10^{-4}, the spatial error of these meshes, while the step count doubles with each decade of tolerance. Capped at BDF2, the controller needs about 30% more steps for the same accuracy. The monolithic step, which adaptive BDF requires, costs about 25% more than the fixed-point step at a fixed Δt (31 against 25 s at $r = 4$, $\Delta t = 0.1$ s).

6 Discussion

Where the time goes. The two codes spend their effort in different places:

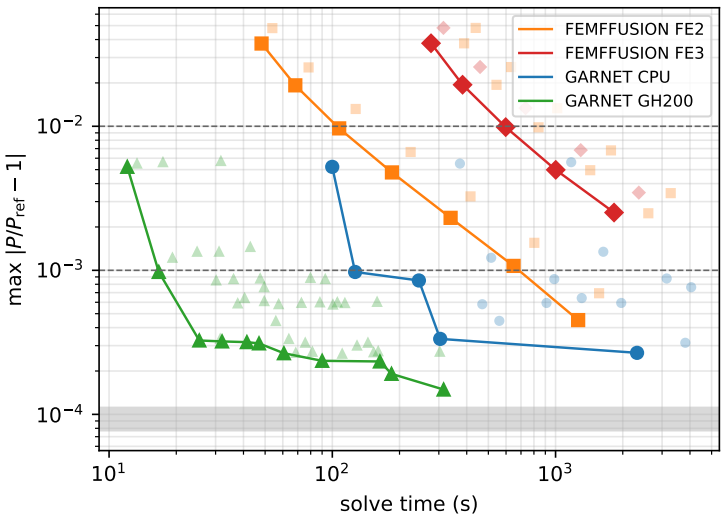


Figure 4: HP-MR transient: largest relative power error over the run against solve time. Each code is scored at its own output times against a spline of the dense reference. Faint: every run; line: each platform's Pareto front.

	`FEMFFUSION`	`GARNET`
Value of one unknown	high: degree-3 elements	low: second-order finite volumes
Outer iterations to converge	4–45 (Krylov eigensolvers)	28–57 (2-group, power iteration); 10 (HP-MR, Wielandt shift)
Solve cost per unknown, $\geq 10^6$ unknowns	36–325 μs (one core)	7.5–64 μs (one core); 0.09–0.9 μs (GH200)
Cost per unknown and inner iteration	–	5–32 ns (one core); 0.03–0.43 ns (GH200)
Memory per unknown	2–3.6 kB (process)	0.2–1.4 kB (CPU); 0.24–2.2 kB (GH200 pool)

- **Steady, one core:** `FEMFFUSION`'s richer unknowns and stronger eigensolver outweigh `GARNET`'s cheaper unknowns, by one to two orders of magnitude in unknowns at the same error.
- **Steady, GH200:** a `GARNET` solve there is 190–320 times faster than on one core from 4 million unknowns up (26–110 times at 1 to 3 million, where the GPU is not yet full). That carries it past `FEMFFUSION` on one core on all three steady cores at 10 pcm and 1%, IAEA-3D included, where it needs 127 million unknowns.
- **Transient:** time integration decides the comparison, not space. `GARNET`'s second-order BDF against `FEMFFUSION`'s first-order semi-implicit schemes is worth a factor of 10 on the same core at 10^{-3}.

What would change the picture.

- **A parallel `FEMFFUSION`.** Its MPI path, once repaired (Section 1), would divide its times by up to the parallel efficiency times the number of cores. At 64 cores and 50% efficiency, a factor of 32 would bring its HP-MR time at 10 pcm from 89 s to about 3 s, close to the GH200's 2.4 s. This is an estimate, not a measurement: the repair is deferred.
- **`GARNET`'s outer acceleration.** The Wielandt shift written for this comparison took the HP-MR from about 250 Anderson-accelerated power iterations to 10 shifted solves, 2.2–3.5 times faster (Table 3). Its shifted solves are preconditioned by a fine-mesh group sweep only. `GARNET` still has no CMFD-style coarse-mesh correction, which would cut the FGMRES iterations within each solve. On the two-group cores the power iteration remains its fastest option, at 28–57 outers against `FEMFFUSION`'s 4–8.

- **GARNET's discretization.** A higher-order or nodal-like discretization would close the steady gap per core. The node powers are where the finite volumes lose the most: on IAEA they need cells well below the 2 cm thermal diffusion length before they converge at second order.
- **FEMFFUSION's time integration.** A working Implicit-Exponential scheme, or any second-order scheme, would shrink the transient gap. Its default scheme is first order and, on this input, does not hold equilibrium.

FEMFFUSION findings. These were found while posing identical problems, and could be reported upstream:

1. Implicit-Exponential takes material 0's decay constants and delayed spectra for the whole core. With multi-material kinetics it also drifts from equilibrium in proportion to Δt (3.9% in a 0.1 s step).
2. An XML material's prompt spectrum stays zero unless the material has a `ChiD` element. With kinetics from `PREC_Filename`, the Implicit-Exponential transient collapses.
3. `full_allocated` with `gd` or `bifpam` gives a wrong k on the refined Langenbuch core: 1.70 and 1.38 instead of 0.9995.
4. `full_matrixfree` with `gd` or `slepc_2g` crashes.
5. Running on more than one MPI rank is not possible yet (Section 1).

Items 1–2 are close to open fix branches of the `ondrejch` fork (`fix/time-delayed-spectrum-implicit`, `fix/xml-default-delayed-spectra`).

Fairness and scope.

- Both codes ran the same problems on the same CPU type, each in its best measured configuration, at tolerances chosen by the same rule.
- The GH200 is compared with one EPYC core. That is FEMFFUSION's only working mode today, not a node-to-node comparison.
- The steady 1 pcm, 0.1% target sits within a factor of three of the references' resolution (0.3–0.4 pcm). It ranks the platforms only coarsely.
- The HP-MR is drums-withdrawn. The inserted, area-smeared drums are a near-black absorber that neither code resolves (Section 2).

A Reproduction

All commands are run from the GARNET repository root on a merlin7 login node, unless stated otherwise.

1. Build FEMFFUSION. The tarballs `petsc-3.19.6`, `slepc-3.19.2` and `dealii-9.5.2` must be in `/data/scratch/shared/poncet_m/femffusion-deps-v1/src`. Then:

```
sbatch slurm/build_femffusion.sh
# or, on a login node: BUILD_JOBS=6 bash -l slurm/build_femffusion.sh
```

This builds the libraries and `~/FEMFFUSION/build-release/femffusion.exe`, writes `build-info.json` (versions, flags and the runtime library path) and runs `femffusion.exe -t`.

2. HP-MR input for `FEMFFUSION`.

```
PYTHONPATH=. python tools/export_femffusion.py \
    /data/scratch/shared/poncet_m/garnet-femffusion-perf/inputs/hpmr-v2
```

The output is the hexagonal mesh, the XML materials (kinetics included), the steady `.prm` and the transient `.prm` with its final-state XML. The steady runs used `hpmr-v1`, written before the kinetics moved into the XML. Its static data are identical, and both inputs give the same k.

3. Runs. Every run is one line of a plan file in `benchmark-results/garnet-vs-femffusion/plans/`: `<chain> <tag> <tool> <arguments>`, where the tool is `tools/assess_garnet_vs_femffusion.py` or `tools/run_femffusion.py`.

```
bash slurm/stage_and_submit_femffusion_cmp.sh cpu-hourly plans/cpu-steady-a.plan
bash slurm/stage_and_submit_femffusion_cmp.sh cpu-daily  plans/cpu-steady-b.plan
bash slurm/stage_and_submit_femffusion_cmp.sh gpu        plans/gpu-steady.plan
```

Each submission stages its own snapshot of the repository with a `PROVENANCE.txt` (`GARNET` commit, status and diff; `FEMFFUSION` commit and build). `slurm/run_perf_plan.sh` runs the chains, pinning each chain on a CPU node to its own NUMA domain. The plans used, in order:

- tuning: `cpu-ff-tune`, `cpu-ff-trtune`, `cpu-garnet-tune`, `gpu-garnet-tune`;
- references: `gpu-ref`, `cpu-ff-ref`, and `cpu-ref-shared` (target `cpu-daily-shared`: part of a shared node, for runs whose time is not used);
- steady: `gpu-steady`, `cpu-steady-a/b`, `cpu-alone`;
- transient: `gpu-transient`, `gpu-transient-2`, `cpu-transient-a/b`, `cpu-fftransient-a/b/c`;
- runs left over by the queue limits: `cpu-split-h1/h2/h3`, `cpu-split-d` and the single-run plans `cpu-one-*`, each an exclusive node with a short time limit.
- Wielandt shift, HP-MR steady: `gpu-wielandt` (`gpu-hourly`) and `cpu-wielandt` (`cpu-hourly`). Both run the fixed and the adaptive shift; `WIELANDT_MODE` in `make_figures.py` selects the rows. The `tune-floor-*` runs of `cpu-wielandt` are the iteration-error check of Section 3. `cpu-one-hpmr-wf-r8` replaced the power-iteration plan `cpu-one-hpmr-r8`, cancelled before it ran. `gpu-2g-wielandt` (`gpu-hourly`) checks the two two-group cores at Langenbuch $r = 12$ and IAEA $r = 24$ (tag `tune-`, not rows).
- IAEA-3D fine meshes: `gpu-iaea-fine` (`gpu-hourly`), $r = 20$, 24, 28, with an $r = 16$ control (tag `tune-`, not a row) that reproduces the `gpu-steady` $r = 16$ run bit for bit on the current code.

4. Report.

```
cd benchmark-results/garnet-vs-femffusion
bash collect.sh                      # results -> data/perf/<job>/
PYTHONPATH=../.. python make_reference.py
make PYTHON=python                   # figures, tables, data/summary.json, PDF
```

Checks in the test suite.

- `tests/validation/test_femffusion_equivalence.py`: `GARNET`'s IAEA-3D and Langenbuch builders pose `FEMFFUSION`'s example inputs, node by node.
- `tests/verification/test_femffusion_io.py`: the XML writer round trip and its kinetics layout.
- `tests/validation/test_hpmr.py`: the hex-homogenized drum option (`absorber="smeared"`).

B All runs

Every comparison run, with its accuracy against the reference of Table 1 and its cost. "Outers / inners" are each code's own counts: `GARNET`'s outer iterations and Krylov iterations, summed over groups, and `FEMFFUSION`'s outer iterations and matrix-vector products. On `GARNET`'s Wielandt-shifted HP-MR rows the FGMRES iterations of the shifted solves are in parentheses. The counts are not comparable across codes. Memory is the CuPy pool for the GH200 and the peak process RSS otherwise.

Platform	Mesh	Unknowns	k	Δk (pcm)	max ΔP	RMS ΔP	Outers / inners	Solve (s)	Wall (s)	Memory (GB)
FEMFFUSION FE2	FE2 r0	80,574	1.0303390	+126.61	26.32%	9.39%	4 / 40 mv	3.02	6.44	0.3
FEMFFUSION FE2	FE2 r1	614,922	1.0292000	+12.71	4.55%	1.64%	4 / 40 mv	41.6	49.5	1.3
FEMFFUSION FE2	FE2 r2	4,803,282	1.0290810	+0.81	0.55%	0.20%	4 / 40 mv	658	718	9.3
FEMFFUSION FE2	FE2 r3	37,967,010	1.0290740	+0.11	0.08%	0.03%	4 / 40 mv	10822	12722	73.0
FEMFFUSION FE3	FE3 r0	263,552	1.0291380	+6.51	5.33%	1.83%	4 / 40 mv	42.6	52	1.1
FEMFFUSION FE3	FE3 r1	2,042,630	1.0290740	+0.11	0.39%	0.14%	4 / 40 mv	665	717	7.3
GARNET CPU	r1	9,158	1.0322967	+322.38	33.47%	10.69%	50 / 152	0.0522	0.0662	0.1
GARNET CPU	r2	73,264	1.0290583	-1.46	31.02%	11.98%	57 / 274	0.346	0.353	0.1
GARNET CPU	r3	247,266	1.0286554	-41.75	25.01%	9.48%	51 / 342	1.29	1.31	0.2
GARNET CPU	r4	586,112	1.0286618	-41.11	19.17%	7.17%	52 / 450	4.25	4.28	0.2
GARNET CPU	r6	1,978,128	1.0287990	-27.38	11.45%	4.22%	52 / 680	36.2	36.4	0.6
GARNET CPU	r8	4,688,896	1.0288949	-17.80	7.30%	2.68%	55 / 937	120	120	1.4
GARNET CPU	r12	15,825,024	1.0289854	-8.75	3.60%	1.31%	48 / 1292	652	654	4.1
GARNET CPU	r16	37,511,168	1.0290220	-5.08	2.08%	0.76%	52 / 1883	2120	2125	9.5
GARNET GH200	r1	9,158	1.0322967	+322.39	33.46%	10.69%	54 / 146	0.138	0.379	0.0
GARNET GH200	r2	73,264	1.0290586	-1.43	31.02%	11.98%	55 / 244	0.177	0.221	0.0
GARNET GH200	r3	247,266	1.0286556	-41.73	25.02%	9.48%	48 / 301	0.185	0.231	0.1
GARNET GH200	r4	586,112	1.0286615	-41.13	19.16%	7.17%	46 / 404	0.23	0.286	0.2
GARNET GH200	r6	1,978,128	1.0287990	-27.39	11.44%	4.22%	48 / 675	0.334	0.399	0.6
GARNET GH200	r8	4,688,896	1.0288948	-17.80	7.30%	2.68%	48 / 934	0.556	0.657	1.4
GARNET GH200	r12	15,825,024	1.0289853	-8.76	3.59%	1.31%	53 / 1584	2.46	2.76	4.9
GARNET GH200	r16	37,511,168	1.0290222	-5.06	2.11%	0.76%	55 / 2270	7.63	8.4	11.4
GARNET GH200	r20	73,264,000	1.0290401	-3.27	1.35%	0.49%	50 / 2598	16.2	17.8	22.9
GARNET GH200	r24	126,600,192	1.0290502	-2.26	0.95%	0.34%	51 / 3197	33.5	36.5	39.6
GARNET GH200	r28	201,036,416	1.0290562	-1.67	0.70%	0.25%	53 / 3878	63.7	69.1	62.9

Table 4: IAEA-3D.

Platform	Mesh	Unknowns	k	Δk (pcm)	max ΔP	RMS ΔP	Outers / inners	Solve (s)	Wall (s)	Memory (GB)
FEMFFUSION FE2	FE2 r0	21,546	1.0002280	+74.42	7.80%	2.90%	8 / 41 mv	0.695	3.65	0.2
FEMFFUSION FE2	FE2 r1	160,802	0.9996945	+21.07	2.15%	0.80%	6 / 31 mv	4.69	10.1	0.5
FEMFFUSION FE2	FE2 r2	1,241,730	0.9995153	+3.15	0.33%	0.12%	5 / 26 mv	44.9	58.5	2.7
FEMFFUSION FE2	FE2 r3	9,758,210	0.9994879	+0.41	0.05%	0.02%	4 / 21 mv	472	576	20.0
FEMFFUSION FE3	FE3 r0	69,440	0.9997480	+26.42	2.70%	1.02%	7 / 36 mv	4.39	8.7	0.4
FEMFFUSION FE3	FE3 r1	530,090	0.9995106	+2.68	0.28%	0.10%	6 / 31 mv	46.3	61.6	2.0
FEMFFUSION FE3	FE3 r2	4,141,346	0.9994865	+0.27	0.03%	0.01%	4 / 21 mv	434	551	14.8
GARNET CPU	r1	2,340	1.0053465	+586.27	39.91%	16.59%	36 / 140	0.0202	0.033	0.1
GARNET CPU	r2	18,720	1.0005438	+106.00	4.13%	2.20%	32 / 194	0.0478	0.0503	0.1
GARNET CPU	r3	63,180	0.9996335	+14.97	1.54%	0.48%	33 / 294	0.152	0.156	0.1
GARNET CPU	r4	149,760	0.9993945	-8.93	2.62%	0.88%	29 / 346	0.384	0.393	0.1
GARNET CPU	r6	505,440	0.9993304	-15.34	2.32%	0.82%	29 / 540	2	2.02	0.2
GARNET CPU	r8	1,198,080	0.9993591	-12.47	1.70%	0.61%	32 / 799	9.03	9.1	0.4
GARNET CPU	r12	4,043,520	0.9994125	-7.13	0.92%	0.33%	32 / 1238	66.1	66.4	1.0
GARNET CPU	r16	9,584,640	0.9994402	-4.36	0.56%	0.20%	32 / 1696	269	270	2.1
GARNET GH200	r1	2,340	1.0053465	+586.27	39.91%	16.58%	34 / 85	0.0902	0.127	0.0
GARNET GH200	r2	18,720	1.0005437	+105.99	4.13%	2.20%	31 / 104	0.0988	0.137	0.0
GARNET GH200	r3	63,180	0.9996334	+14.96	1.54%	0.48%	32 / 164	0.119	0.155	0.0
GARNET GH200	r4	149,760	0.9993944	-8.94	2.62%	0.88%	31 / 207	0.131	0.178	0.0
GARNET GH200	r6	505,440	0.9993303	-15.35	2.32%	0.82%	32 / 329	0.178	0.238	0.1
GARNET GH200	r8	1,198,080	0.9993592	-12.46	1.69%	0.61%	28 / 405	0.207	0.277	0.3
GARNET GH200	r12	4,043,520	0.9994125	-7.12	0.92%	0.33%	32 / 714	0.35	0.449	1.0
GARNET GH200	r16	9,584,640	0.9994403	-4.35	0.56%	0.20%	32 / 966	0.85	1.04	2.4

Table 5: 3-D Langenbuch at $t = 0$.

Platform	Mesh	Unknowns	k	Δk (pcm)	max ΔP	RMS ΔP	Outers / inners	Solve (s)	Wall (s)	Memory (GB)
FEMFFUSION FE2	FE2 r0	165,165	1.1188600	+22.00	0.58%	0.36%	45 / 46 mv	17.5	21.3	0.6
FEMFFUSION FE2	FE2 r1	1,239,799	1.1186320	-0.80	0.09%	0.05%	33 / 35 mv	124	132	3.3
FEMFFUSION FE2	FE2 r2	9,602,307	1.1186360	-0.40	0.01%	0.00%	20 / 24 mv	967	1025	24.4
FEMFFUSION FE3	FE3 r0	534,347	1.1186060	-3.40	0.14%	0.08%	11 / 56 mv	89.3	97.5	1.8
FEMFFUSION FE3	FE3 r1	4,095,113	1.1186360	-0.40	0.01%	0.00%	7 / 36 mv	675	730	13.2
GARNET CPU	r2 z30	435,600	1.1178234	-81.66	3.81%	1.90%	10 (61) / 3194	8.27	8.41	0.7
GARNET CPU	r3 z40	1,306,800	1.1182428	-39.72	2.17%	1.09%	10 (59) / 4556	31	31.3	1.9
GARNET CPU	r4 z50	2,904,000	1.1184069	-23.31	1.40%	0.71%	10 (60) / 6225	94.7	95.3	3.9
GARNET CPU	r6 z80	10,454,400	1.1185564	-8.36	0.59%	0.29%	10 (61) / 9837	667	670	12.3
GARNET CPU	r8 z100	23,232,000	1.1185921	-4.79	0.37%	0.19%	10 (61) / 13335	1835	1843	23.6
GARNET GH200	r2 z30	435,600	1.1178234	-81.66	3.81%	1.90%	10 (58) / 1793	0.977	1.36	1.1
GARNET GH200	r3 z40	1,306,800	1.1182428	-39.72	2.17%	1.09%	10 (57) / 2392	1.17	1.63	2.6
GARNET GH200	r4 z50	2,904,000	1.1184069	-23.31	1.40%	0.71%	10 (57) / 3127	1.36	1.89	5.7
GARNET GH200	r6 z80	10,454,400	1.1185564	-8.36	0.59%	0.29%	10 (59) / 4900	2.38	3.28	20.1
GARNET GH200	r8 z100	23,232,000	1.1185921	-4.79	0.37%	0.19%	10 (60) / 6613	5.23	7	50.1
GARNET GH200	r12 z160	83,635,200	1.1186254	-1.46	0.15%	0.08%	10 (61) / 10481	23.2	30.6	85.3

Table 6: HP-MR, steady.

Table 7: HP-MR transient. Errors: the largest $|P/P_{\mathrm{ref}} - 1|$ over the run (max) and at $t = 2$ s (end). Schemes: `FEMFFUSION`'s Semi-Implicit-Euler (SI-Euler) and Semi-Implicit-Exponential (SI-Exp); `GARNET`'s fixed-step BDF2 with the fixed-point step, or with the monolithic step (mono); aBDFq (ϵ): adaptive step with automatic order up to q and per-step error tolerance ϵ, with Δt the range of accepted steps. Steps: the accepted time steps of the march (rejected ones in parentheses). Inners: its inner iterations, `GARNET`'s Krylov iterations and `FEMFFUSION`'s GMRES iterations (one linear solve per step, plus a null step at $t = 0$).

Mesh	Unknowns	Scheme	Δt (s)	max err	end err	Steps	Inners	Solve (s)	Wall (s)	Mem. (GB)
FEMFFUSION FE2										
FE2 r0	165,165	SI-Euler	0.2	4.8e-02	4.8e-02	10	51	53.9	76.1	1.9
FE2 r0	165,165	SI-Euler	0.1	2.6e-02	2.6e-02	20	63	78.3	85.9	1.9
FE2 r0	165,165	SI-Euler	0.05	1.3e-02	1.3e-02	40	84	127	138	1.9
FE2 r0	165,165	SI-Euler	0.025	6.6e-03	6.6e-03	80	122	224	240	1.9
FE2 r0	165,165	SI-Euler	0.0125	3.3e-03	3.3e-03	160	193	416	459	1.9
FE2 r0	165,165	SI-Euler	0.00625	1.6e-03	1.6e-03	320	349	803	865	1.9
FE2 r0	165,165	SI-Euler	0.003125	6.9e-04	6.9e-04	640	664	1567	1681	1.9
FE2 r0	165,165	SI-Exp	0.2	3.8e-02	3.8e-02	10	51	48.2	70	1.6
FE2 r0	165,165	SI-Exp	0.1	1.9e-02	1.9e-02	20	65	68.3	75.6	1.6
FE2 r0	165,165	SI-Exp	0.05	9.7e-03	9.7e-03	40	85	107	117	1.6
FE2 r0	165,165	SI-Exp	0.025	4.8e-03	4.8e-03	80	121	185	201	1.6
FE2 r0	165,165	SI-Exp	0.0125	2.3e-03	2.3e-03	160	193	339	381	1.6
FE2 r0	165,165	SI-Exp	0.00625	1.1e-03	1.1e-03	320	349	648	711	1.6
FE2 r0	165,165	SI-Exp	0.003125	4.5e-04	4.5e-04	640	664	1262	1367	1.6
FE2 r1	1,239,799	SI-Euler	0.2	4.8e-02	4.8e-02	10	57	439	466	13.5
FE2 r1	1,239,799	SI-Euler	0.1	2.6e-02	2.6e-02	20	68	632	651	13.5
FE2 r1	1,239,799	SI-Euler	0.05	1.3e-02	1.3e-02	40	92	1017	1056	13.5
FE2 r1	1,239,799	SI-Euler	0.025	6.8e-03	6.8e-03	80	130	1771	1821	13.5
FE2 r1	1,239,799	SI-Euler	0.0125	3.4e-03	3.4e-03	160	204	3280	3352	13.5
FE2 r1	1,239,799	SI-Exp	0.2	3.8e-02	3.8e-02	10	57	389	413	11.4
FE2 r1	1,239,799	SI-Exp	0.1	1.9e-02	1.9e-02	20	69	544	559	11.4
FE2 r1	1,239,799	SI-Exp	0.05	9.8e-03	9.8e-03	40	92	839	871	11.4
FE2 r1	1,239,799	SI-Exp	0.025	5.0e-03	5.0e-03	80	127	1430	1470	11.4
FE2 r1	1,239,799	SI-Exp	0.0125	2.5e-03	2.5e-03	160	203	2603	2661	11.4
FEMFFUSION FE3										
FE3 r0	534,347	SI-Euler	0.2	4.8e-02	4.8e-02	10	50	315	345	9.1
FE3 r0	534,347	SI-Euler	0.1	2.6e-02	2.6e-02	20	63	459	477	9.2
FE3 r0	534,347	SI-Euler	0.05	1.3e-02	1.3e-02	40	86	734	773	9.1
FE3 r0	534,347	SI-Euler	0.025	6.8e-03	6.8e-03	80	120	1296	1343	9.1
FE3 r0	534,347	SI-Euler	0.0125	3.5e-03	3.5e-03	160	193	2360	2433	9.1
FE3 r0	534,347	SI-Exp	0.2	3.8e-02	3.8e-02	10	52	277	304	7.5
FE3 r0	534,347	SI-Exp	0.1	1.9e-02	1.9e-02	20	63	383	397	7.5
FE3 r0	534,347	SI-Exp	0.05	9.9e-03	9.9e-03	40	86	598	633	7.5
FE3 r0	534,347	SI-Exp	0.025	5.0e-03	5.0e-03	80	119	1002	1040	7.4
FE3 r0	534,347	SI-Exp	0.0125	2.5e-03	2.5e-03	160	193	1828	1886	7.5
GARNET CPU										
r2 z30	435,600	BDF2	0.2	5.2e-03	5.2e-03	10	36,921	99.9	100	0.8
r2 z30	435,600	BDF2	0.1	9.7e-04	9.4e-04	20	54,912	127	127	0.8
r2 z30	435,600	BDF2	0.05	8.5e-04	2.1e-04	40	120,496	244	244	0.8
r2 z30	435,600	BDF2	0.025	5.8e-04	5.2e-04	80	246,730	470	470	0.8
r2 z30	435,600	BDF2	0.0125	5.9e-04	5.9e-04	160	500,208	911	911	0.9
r2 z30	435,600	aBDF2 (10^{-3})	1.5e-4–0.14	3.3e-04	3.2e-04	33 (6 rej.)	148,333	304	304	1.4

Mesh	Unknowns	Scheme	Δt (s)	max err	end err	Steps	Inners	Solve (s)	Wall (s)	Mem. (GB)
r2 z30	435,600	aBDF2 (10^{-4})	1.7e-5–0.061	4.5e-04	4.5e-04	65 (10 rej.)	274,183	560	560	1.4
r3 z40	1,306,800	BDF2	0.2	5.5e-03	5.5e-03	10	53,960	373	373	2.2
r3 z40	1,306,800	BDF2	0.1	1.2e-03	1.2e-03	20	81,887	515	515	2.2
r3 z40	1,306,800	BDF2	0.05	8.7e-04	6.6e-05	40	173,070	987	987	2.2
r3 z40	1,306,800	BDF2	0.025	5.9e-04	2.4e-04	80	367,015	1969	1969	2.2
r3 z40	1,306,800	BDF2	0.0125	3.1e-04	3.1e-04	160	740,862	3812	3813	2.2
r3 z40	1,306,800	aBDF2 (10^{-3})	1.6e-4–0.15	6.4e-04	6.3e-04	33 (6 rej.)	219,191	1310	1310	3.8
r3 z40	1,306,800	aBDF2 (10^{-4})	1.8e-5–0.061	2.7e-04	1.7e-04	64 (9 rej.)	395,570	2317	2317	3.8
r4 z50	2,904,000	BDF2	0.2	5.6e-03	5.6e-03	10	72,796	1175	1175	4.6
r4 z50	2,904,000	BDF2	0.1	1.3e-03	1.3e-03	20	109,755	1636	1636	4.6
r4 z50	2,904,000	BDF2	0.05	8.8e-04	1.9e-04	40	231,712	3146	3146	4.7
r4 z50	2,904,000	aBDF2 (10^{-3})	1.6e-4–0.15	7.6e-04	7.5e-04	33 (6 rej.)	291,318	4050	4050	8.2
GARNET GH200										
r2 z30	435,600	BDF2	0.2	5.2e-03	5.2e-03	10	22,561	12.1	19	1.0
r2 z30	435,600	BDF2	0.1	9.7e-04	9.4e-04	20	32,902	16.7	22	1.0
r2 z30	435,600	BDF2	0.05	8.5e-04	2.1e-04	40	71,728	30.2	35.3	1.0
r2 z30	435,600	BDF2	0.025	5.8e-04	5.1e-04	80	146,770	57.5	62.2	1.1
r2 z30	435,600	BDF2	0.0125	5.9e-04	5.9e-04	160	298,838	114	118	1.1
r2 z30	435,600	aBDF2 (10^{-3})	1.5e-4–0.14	3.3e-04	3.2e-04	33 (6 rej.)	84,776	31.5	38.3	1.6
r2 z30	435,600	aBDF2 (10^{-4})	1.6e-5–0.061	4.4e-04	4.4e-04	65 (10 rej.)	157,225	55.8	60.4	1.6
r2 z30	435,600	aBDF2 (10^{-5})	1.2e-6–0.028	5.8e-04	5.8e-04	134 (14 rej.)	292,497	101	105	1.6
r2 z30	435,600	aBDF4 (10^{-3})	1.5e-4–0.21	3.3e-04	2.6e-04	26 (6 rej.)	67,247	25.4	29.9	1.6
r2 z30	435,600	aBDF4 (10^{-4})	1.6e-5–0.13	5.9e-04	5.9e-04	42 (11 rej.)	102,393	37.7	42.2	1.6
r2 z30	435,600	aBDF4 (10^{-5})	1.2e-6–0.091	6.1e-04	6.1e-04	64 (14 rej.)	139,013	49.5	54	1.6
r3 z40	1,306,800	BDF2	0.2	5.5e-03	5.5e-03	10	30,530	13.3	19.8	2.8
r3 z40	1,306,800	BDF2	0.1	1.2e-03	1.2e-03	20	46,216	19.2	25.7	2.8
r3 z40	1,306,800	BDF2	0.05	8.7e-04	6.6e-05	40	97,443	36.1	42.5	2.8
r3 z40	1,306,800	BDF2	0.025	5.9e-04	2.4e-04	80	206,263	72.7	79	3.1
r3 z40	1,306,800	BDF2	0.0125	3.1e-04	3.1e-04	160	417,393	144	150	3.1
r3 z40	1,306,800	aBDF2 (10^{-3})	1.6e-4–0.15	6.4e-04	6.3e-04	33 (6 rej.)	119,349	40.5	46.4	4.6
r3 z40	1,306,800	aBDF2 (10^{-4})	1.8e-5–0.061	2.7e-04	1.7e-04	64 (9 rej.)	215,738	68.6	74.4	4.6
r3 z40	1,306,800	aBDF2 (10^{-5})	2.0e-6–0.028	3.0e-04	3.0e-04	133 (13 rej.)	409,449	129	135	4.6
r3 z40	1,306,800	aBDF4 (10^{-3})	1.6e-4–0.21	3.2e-04	2.9e-05	26 (6 rej.)	94,192	32.1	37.8	4.6
r3 z40	1,306,800	aBDF4 (10^{-4})	1.8e-5–0.14	3.1e-04	3.1e-04	41 (11 rej.)	141,453	46.9	52.8	4.6
r3 z40	1,306,800	aBDF4 (10^{-5})	2.0e-6–0.089	3.3e-04	3.3e-04	63 (14 rej.)	193,602	63.9	69.8	4.6
r4 z50	2,904,000	BDF2	0.2	5.6e-03	5.6e-03	10	39,472	17.4	26	6.1
r4 z50	2,904,000	BDF2	0.1	1.3e-03	1.3e-03	20	59,474	24.8	33.4	6.1
r4 z50	2,904,000	BDF2	0.05	8.8e-04	1.9e-04	40	125,912	47.5	56.3	6.1
r4 z50	2,904,000	BDF2	0.025	6.0e-04	1.1e-04	80	266,513	88	96	6.1
r4 z50	2,904,000	BDF2	0.0125	1.9e-04	1.9e-04	160	543,348	184	192	6.1
r4 z50	2,904,000	BDF2 mono	0.1	1.3e-03	1.3e-03	20	94,211	31.1	38.7	9.3
r4 z50	2,904,000	BDF2 mono	0.025	6.0e-04	1.1e-04	80	346,419	105	113	9.3
r4 z50	2,904,000	aBDF2 (10^{-3})	1.6e-4–0.15	7.6e-04	7.5e-04	33 (6 rej.)	155,460	49.5	56.9	9.9
r4 z50	2,904,000	aBDF2 (10^{-4})	1.8e-5–0.06	2.4e-04	4.2e-05	65 (10 rej.)	284,749	90	97.7	9.9
r4 z50	2,904,000	aBDF2 (10^{-5})	2.1e-6–0.028	2.7e-04	1.8e-04	133 (13 rej.)	533,385	160	168	9.9
r4 z50	2,904,000	aBDF4 (10^{-3})	1.6e-4–0.22	3.2e-04	1.8e-04	26 (6 rej.)	123,144	41.4	49	9.9
r4 z50	2,904,000	aBDF4 (10^{-4})	1.8e-5–0.13	2.7e-04	1.9e-04	41 (11 rej.)	185,466	60.7	68.3	9.9
r4 z50	2,904,000	aBDF4 (10^{-5})	2.1e-6–0.092	2.7e-04	2.1e-04	63 (15 rej.)	257,575	81.4	88.9	9.9
r6 z80	10,454,400	BDF2	0.2	5.7e-03	5.7e-03	10	61,933	31.7	46.6	21.5
r6 z80	10,454,400	BDF2	0.1	1.5e-03	1.5e-03	20	88,332	42.9	57.7	21.5

Mesh	Unknowns	Scheme	Δt (s)	max err	end err	Steps	Inners	Solve (s)	Wall (s)	Mem. (GB)
r6 z80	10,454,400	BDF2	0.05	8.9e-04	3.0e-04	40	186,315	79.8	94.5	21.5
r6 z80	10,454,400	BDF2	0.025	6.0e-04	5.9e-06	80	397,409	159	173	21.5
r6 z80	10,454,400	BDF2	0.0125	1.5e-04	8.4e-05	160	810,694	315	329	21.5
r6 z80	10,454,400	aBDF2 (10^{-3})	1.6e-4–0.15	8.7e-04	8.6e-04	33 (6 rej.)	230,345	93	107	34.7
r6 z80	10,454,400	aBDF2 (10^{-4})	1.9e-5–0.06	2.3e-04	6.5e-05	65 (10 rej.)	422,465	163	177	34.7
r6 z80	10,454,400	aBDF2 (10^{-5})	2.1e-6–0.028	2.7e-04	7.1e-05	133 (13 rej.)	796,438	303	317	34.7
r6 z80	10,454,400	aBDF4 (10^{-3})	1.6e-4–0.22	3.1e-04	2.9e-04	26 (6 rej.)	182,028	75.6	89.2	34.7
r6 z80	10,454,400	aBDF4 (10^{-4})	1.9e-5–0.13	2.6e-04	8.5e-05	41 (11 rej.)	275,637	111	124	34.7
r6 z80	10,454,400	aBDF4 (10^{-5})	2.1e-6–0.092	2.7e-04	1.0e-04	63 (15 rej.)	384,542	153	167	34.7